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THERMAL FLOW METER AND FLOW RATE CALCULATION METHOD

Abstract

Provided is a flow rate calculation method including: a time measurement step of measuring a first reach time taken by a liquid heated by a heating resistance element to reach a first temperature detecting resistance element and measuring a second reach time taken by a liquid heated by the heating resistance element to reach the second temperature detecting resistance element; and a calculation step of calculating a liquid flow rate of a liquid flowing through a measurement tube based on the first and second reach time. The calculation step includes calculating the liquid flow rate based on both the first and second reach time when a first flow rate is greater than a first predetermined flow rate and calculating the liquid flow rate based on the first reach time without using the second reach time when the first flow rate is not greater than the first predetermined flow rate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims foreign priority benefits under U.S.C. § 119 to Japanese Patent Application No. 2024-035938 filed on Mar. 8, 2024, the contents of which is hereby incorporated by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a thermal flow meter and a flow rate calculation method.

2. Description of Related Art

[0003] Conventionally, there are thermal flow meters having a temperature detection substrate in which a heating resistance element and a temperature detecting resistance element are formed on a detection surface along an axis and configured to calculate a liquid flow rate based on a time difference between the time of heating the liquid by the heating resistance element and the time of detecting the heated liquid by the temperature detecting resistance element (for example, Japanese Patent Application Laid-Open No. 2021-131323).

[0004] Japanese Patent Application Laid-Open No. 2021-131323 discloses that the control substrate calculates the flow speed of a liquid flowing through a measurement tube based on the time of instantaneously heating a heating resistance element and the time of two temperature detecting resistance elements then determining the temperature of the heated liquid and calculates the liquid flow rate based on the flow speed and the sectional area of the measurement tube.

[0005] The thermal flow meter disclosed in Japanese Patent Application Laid-Open No. 2021-131323 has a first temperature detecting resistance element arranged downstream from the heating resistance element in the flow direction of a liquid and has a second temperature detecting resistance element arranged further downstream from the first temperature detecting resistance element. When the thermal flow meter disclosed in Japanese Patent Application Laid-Open No. 2021-131323 is used, for example, a first flow rate is calculated based on the time of instantaneously heating the heating resistance element and the time of determining the temperature of the heated liquid by the first temperature detecting resistance element, a second flow rate is calculated based on the time of instantaneously heating the heating resistance element and the time of determining the temperature of the heated liquid by the second temperature detecting resistance element, and the liquid flow rate can be calculated by averaging the first flow rate and the second flow rate.

[0006] When the liquid flow rate of a liquid flowing through the measurement tube is relatively large, the temperature difference between the temperature of the heated liquid reaching the first temperature detecting resistance element and the temperature of the heated liquid reaching the second temperature detecting resistance element is small, and the first flow rate and the second flow rate are substantially the same values. Thus, it is possible to suitably calculate the liquid flow rate by averaging the first flow rate and the second flow rate.

[0007] When the liquid flow rate of a liquid flowing through the measurement tube is relatively small, the temperature of the heated liquid reaching the second temperature detecting resistance element is lower than the temperature of the heated liquid reaching the first temperature detecting resistance element, and it is not possible to suitably calculate the liquid flow rate based on the first flow rate and the second flow rate. This is because, when the liquid flow rate is relatively small, the

heat applied to the liquid from the heating resistance element is transferred to the measurement tube or the liquid itself before reaching the second temperature detecting resistance element.

[0008] The present disclosure has been made in view of such circumstances and intends to provide a thermal flow meter and a flow rate calculation method that can reduce a calculation error in the flow rate occurring when the liquid flow rate is relatively small and thereby expand the measurable flow rate range on the lower limit value side.

BRIEF SUMMARY

[0009] The present disclosure employs the following solutions in order to achieve the above object.

[0010] A thermal flow meter according to one aspect of the present disclosure includes: a measurement tube having an inlet and an outlet, a liquid flowing in the inlet and the liquid, which flowed in from the inlet, flowing out of the outlet, and having an internal flow passage extending along an axis formed in the measurement tube; a temperature detection substrate having a heating resistance element, a first temperature detecting resistance element, and a second temperature detecting resistance element that are formed on a detection surface, each extend in a flow direction from the inlet to the outlet, and are spaced apart from each other; a voltage output unit configured to output a voltage signal to the temperature detection substrate, the voltage signal being for heating the heating resistance element; a voltage detection unit configured to determine a first output voltage in accordance with a temperature of the first temperature detecting resistance element and a second output voltage in accordance with a temperature of the second temperature detecting resistance element; a time measurement unit configured to measure a first reach time based on the voltage signal and the first output voltage and measure a second reach time based on the voltage signal and the second output voltage, the first reach time being a time taken by a liquid heated by the heating resistance element to reach the first temperature detecting resistance element, and the second reach time being a time taken by a liquid heated by the heating resistance element to reach the second temperature detecting resistance element; and a calculation unit configured to calculate a liquid flow rate of a liquid flowing through the measurement tube based on the first reach time and the second reach time, and the calculation unit calculates the liquid flow rate based on both the first reach time and the second reach time when the liquid flow rate is greater than a first predetermined flow rate, and the calculation unit calculates the liquid flow rate based on the first reach time without using the second reach time when the liquid flow rate is less than or equal to the first predetermined flow rate.

[0011] According to the thermal flow meter of one aspect of the present disclosure, when the liquid flow rate is greater than the first predetermined flow rate, the liquid flow rate is calculated based on both the first reach time and the second reach time. Thus, the error in calculation of the flow rate can be reduced compared to a case where the flow rate is calculated without the use of either one of the first reach time and the second reach time.

[0012] Further, according to the thermal flow meter of one aspect of the present disclosure, when the liquid flow rate is less than or equal to the first predetermined flow rate, the liquid flow rate is calculated based on the first reach time without the use of the second reach time. Thus, the calculation error of the liquid flow rate due to heat applied to a liquid from the heating resistance element being transferred to the measurement tube or the liquid itself before reaching the second temperature detecting resistance element can be reduced. Therefore, according to the thermal flow meter of one aspect of the present disclosure, the calculation error in the flow rate occurring when the liquid flow rate is relatively small can be reduced, and thereby the measurable flow rate range can be expanded on the lower limit value side.

[0013] The thermal flow meter according to one aspect of the present disclosure may be configured such that the calculation unit calculates a liquid flow rate by applying an equal weight to a first flow rate calculated based on the first reach time and to a second flow rate calculated based on the second reach time when the liquid flow rate is greater than a second predetermined flow rate, which is greater than the first predetermined flow rate, and the calculation unit calculates a liquid

flow rate by increasing a weight on the first flow rate to be greater than a weight on the second flow rate when the liquid flow rate is greater than the first predetermined flow rate and less than or equal to the second predetermined flow rate.

[0014] According to the thermal flow meter of the present configuration, when the liquid flow rate is greater than the second predetermined flow rate, the liquid flow rate is calculated with an equal weight being applied to the first flow rate calculated based on the first reach time and the second flow rate calculated based on the second reach time. Thus, the error in calculation of the flow rate can be reduced compared to a case where the flow rate is calculated based only on either one of the first flow rate calculated based on the first reach time and the second flow rate calculated based on the second reach time.

[0015] Further, according to the thermal flow meter of the present configuration, when the liquid flow rate is greater than the first predetermined flow rate and less than or equal to the second predetermined flow rate, the liquid flow rate is calculated with the weight on the first flow rate being increased to be greater than the weight on the second flow rate. Thus, the calculation error of the liquid flow rate due to heat applied to a liquid from the heating resistance element being transferred to the measurement tube or the liquid itself before reaching the second temperature detecting resistance element can be reduced.

[0016] The thermal flow meter of the above configuration may be formed such that, as the liquid flow rate calculated by the calculation unit decreases from the second predetermined flow rate to the first predetermined flow rate, the calculation unit calculates the liquid flow rate by gradually increasing the weight on the first flow rate at a constant increase rate and by gradually reducing the weight on the second flow rate at a constant reduction rate.

[0017] According to the thermal flow meter of the present form, since the weight on the first flow rate is gradually increased at a constant increase rate and the weight on the second flow rate is gradually reduced at a constant reduction rate as the liquid flow rate calculated by the calculation unit decreases and is closer to the first predetermined flow rate from the second predetermined flow rate, fluctuation in the calculation result of the flow rate can be suitably suppressed.

[0018] The thermal flow meter according to one aspect of the present disclosure may be configured such that the calculation unit calculates the liquid flow rate by applying predetermined weights to a first flow rate calculated based on the first reach time and to a second flow rate calculated based on the second reach time when a liquid flow rate is greater than the first predetermined flow rate, and the calculation unit calculates the liquid flow rate by increasing a weight on the first flow rate to be greater than the predetermined weights when the liquid flow rate is less than or equal to the first predetermined flow rate.

[0019] According to the thermal flow meter of the present configuration, when the liquid flow rate is less than or equal to the first predetermined flow rate, the liquid flow rate is calculated with the weight on the first flow rate being increased to be greater than the predetermined weight. Thus, the calculation error of the liquid flow rate due to heat applied to a liquid from the heating resistance element being transferred to the measurement tube or the liquid itself before reaching the second temperature detecting resistance element can be reduced. Therefore, according to the thermal flow meter of one aspect of the present disclosure, the calculation error in the flow rate occurring when the liquid flow rate is relatively small can be reduced, and thereby the measurable flow rate range can be expanded on the lower limit value side.

[0020] A flow rate calculation method according to one aspect of the present disclosure is a flow rate calculation method for calculating a liquid flow rate by a thermal flow meter, the thermal flow meter has a measurement tube having an inlet and an outlet, a liquid flowing in the inlet and the liquid, which flowed in from the inlet, flowing out of the outlet, and having an internal flow passage extending along an axis formed in the measurement tube, a temperature detection substrate having a heating resistance element, a first temperature detecting resistance element, and a second temperature detecting resistance element that are formed on a detection surface, each extend in a

flow direction from the inlet to the outlet, and are spaced apart from each other, a voltage output unit configured to output a voltage signal to the temperature detection substrate, the voltage signal being for heating the heating resistance element, and a voltage detection unit configured to determine a first output voltage in accordance with a temperature of the first temperature detecting resistance element and a second output voltage in accordance with a temperature of the second temperature detecting resistance element, the flow rate calculation method includes: a time measurement step of measuring a first reach time based on the voltage signal and the first output voltage and measuring a second reach time based on the voltage signal and the second output voltage, the first reach time being a time taken by a liquid heated by the heating resistance element to reach the first temperature detecting resistance element, and the second reach time being a time taken by a liquid heated by the heating resistance element to reach the second temperature detecting resistance element; and a calculation step of calculating a liquid flow rate of a liquid flowing through the measurement tube based on the first reach time and the second reach time, and the calculation step includes calculating the liquid flow rate based on both the first reach time and the second reach time when the liquid flow rate is greater than a first predetermined flow rate and calculating the liquid flow rate based on the first reach time without using the second reach time when the liquid flow rate is less than or equal to the first predetermined flow rate.

[0021] According to the flow rate calculation method of one aspect of the present disclosure, when the liquid flow rate is greater than the first predetermined flow rate, the liquid flow rate is calculated based on both the first reach time and the second reach time. Thus, the error in calculation of the flow rate can be reduced compared to a case where the flow rate is calculated without the use of either one of the first reach time and the second reach time.

[0022] Further, according to the flow rate calculation method according to one aspect of the present disclosure, when the liquid flow rate is less than or equal to the first predetermined flow rate, the liquid flow rate is calculated based on the first reach time without the use of the second reach time. Thus, the calculation error of the liquid flow rate due to heat applied to a liquid from the heating resistance element being transferred to the measurement tube or the liquid itself before reaching the second temperature detecting resistance element can be reduced. Therefore, according to the flow rate calculation method of one aspect of the present disclosure, the calculation error in the flow rate occurring when the liquid flow rate is relatively small can be reduced.

[0023] The flow rate calculation method according to one aspect of the present disclosure may be configured such that the calculation step includes calculating the liquid flow rate by applying predetermined weights to a first flow rate calculated based on the first reach time and to a second flow rate calculated based on the second reach time when a liquid flow rate is greater than the first predetermined flow rate and calculating the liquid flow rate by increasing a weight on the first flow rate to be greater than the predetermined weights when the liquid flow rate is less than or equal to the first predetermined flow rate.

[0024] According to the flow rate calculation method of the present configuration, when the liquid flow rate is less than or equal to the first predetermined flow rate, the liquid flow rate is calculated with the weight on the first flow rate being increased to be greater than the predetermined weight. Thus, the calculation error of the liquid flow rate due to heat applied to a liquid from the heating resistance element being transferred to the measurement tube or the liquid itself before reaching the second temperature detecting resistance element can be reduced. Therefore, according to the flow rate calculation method of one aspect of the present disclosure, the calculation error in the flow rate occurring when the liquid flow rate is relatively small can be reduced.

[0025] According to the present disclosure, it is possible to provide a thermal flow meter and a flow rate calculation method that can reduce a calculation error in the flow rate occurring when the liquid flow rate is relatively small and thereby expand the measurable flow rate range on the lower limit value side.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0026] FIG. 1 is a longitudinal sectional view of a thermal flow meter according to one embodiment of the present disclosure.

[0027] FIG. 2 is a partial enlarged view of a sensor unit illustrated in FIG. 1.

[0028] FIG. 3 is an arrow A-A end face view of the sensor unit illustrated in FIG. 2.

[0029] FIG. 4 is a plan view of a sensor substrate illustrated in FIG. 3 when viewed from the detection surface side.

[0030] FIG. 5 is a block diagram illustrating a control configuration of a control substrate illustrated in FIG. 1.

[0031] FIG. 6 is a graph illustrating a relationship between the liquid flow rate calculated by a calculation unit and amplitudes of a first differential voltage and a second differential voltage.

[0032] FIG. 7 is a flowchart illustrating a flow rate calculation method for calculating a liquid flow rate performed by the thermal flow meter according to one embodiment of the present disclosure.

[0033] FIG. 8 is a graph illustrating a relationship between the liquid flow rate calculated by the calculation unit and a first coefficient and a second coefficient used in calculation formulae.

[0034] FIG. 9 is a graph illustrating a modified example of the relationship between the liquid flow rate calculated by the calculation unit and the first coefficient and the second coefficient used in the calculation formulae.

DETAILED DESCRIPTION

[0035] A thermal flow meter **100** according to one embodiment of the present disclosure will be described below with reference to the drawings.

[0036] FIG. 1 is a longitudinal sectional view of the thermal flow meter **100** according to one embodiment of the present disclosure.

[0037] The thermal flow meter **100** of the present embodiment is a thermal flow meter that heats a liquid flowing through an internal flow passage, determines the temperature of the heated liquid, and thereby measures the liquid flow rate. The thermal flow meter **100** of the present embodiment is suitable for measuring a micro-flow rate from 4 $\mu\text{L}/\text{min}$ to 350 $\mu\text{L}/\text{min}$, for example. As illustrated in FIG. 1, the thermal flow meter **100** of the present embodiment has a sensor unit **10**, a control substrate **20**, a relay substrate **30**, an upper case **40**, and a bottom case **50**.

[0038] As illustrated in FIG. 1, the sensor unit **10** causes a liquid flowing in from an inlet **10a** connected to an external pipe (not illustrated) to flow out of an outlet **10b** connected to an external pipe (not illustrated), heats the liquid flowing through an internal flow passage **10c**, and outputs a voltage in accordance with the temperature of the heated liquid to the control substrate **20**. The sensor unit **10** will be described later in detail.

[0039] The control substrate **20** is a device that outputs a voltage signal used for heating a liquid to the sensor unit **10**, receives a voltage signal in accordance with the temperature of the heated liquid from the sensor unit **10**, and calculates the liquid flow rate of the liquid flowing through the sensor unit **10**. The control substrate **20** will be described later in detail.

[0040] The relay substrate **30** relays various signals transmitted and received between the control substrate **20** and an external device (not shown). A cable **200** for transmitting and receiving the various signals to and from the external device (not shown) is connected to the relay substrate **30**.

[0041] The upper case **40** serves as a housing for an upper portion of the thermal flow meter **100**, and accommodates the control substrate **20** inside.

[0042] The bottom case **50** serves as a housing for a lower portion of the thermal flow meter **100**, and accommodates the sensor unit **10** inside. With the sensor unit **10** inserted in the bottom case **50**, a stopper **70** is inserted between the bottom case **50** and the sensor unit **10** from the inlet **10a** side

of the sensor unit **10**.

[0043] Similarly, with the sensor unit **10** inserted in the bottom case **50**, the stopper **70** is inserted between the bottom case **50** and the sensor unit **10** from the outlet **10b** side of the sensor unit **10**. The sensor unit **10** becomes fixed to the bottom case **50** by means of the stopper **70**.

[0044] The bottom case **50** has fastening holes **50a** on its bottom surface and is fixed to an installation surface (not shown) by fastening bolts (not shown) that are inserted from below the installation surface.

[0045] Next, the sensor unit **10** will be described in detail with reference to FIGS. **2** to **4**. FIG. **2** is a partial enlarged view of a sensor unit illustrated in FIG. **1**. FIG. **3** is an arrow A-A end face view of the sensor unit illustrated in FIG. **2**. FIG. **4** is a plan view of a sensor substrate illustrated in FIG. **3** when viewed from the detection surface side.

[0046] As shown in FIG. **2**, the sensor unit **10** has a measurement tube **11**, a sensor substrate **12** (temperature detecting substrate), a nut **15**, an inlet-side body **16**, and an outlet-side body **17**.

[0047] The measurement tube **11** is a tube including an inlet **11a** through which liquid enters and an outlet **11b** through which the liquid flowing from the inlet **11a** exits. As shown in FIG. **3**, the measurement tube **11** includes the internal flow passage **10c** extending along an axis X and having a circular cross section. The measurement tube **11** is, for example, made of glass (such as quartz glass with high silicon dioxide content). Furthermore, the measurement tube **11** may be a tube element having the corrosion resistance to alkaline liquids.

[0048] As shown in FIG. **4**, the sensor substrate **12** is a substrate made of glass (e.g., made of quartz glass having a high silicon dioxide content) with the temperature detecting resistance wire **12e**, the temperature detecting resistance wire **12c**, the heating resistance wire (the heating resistance element) **12a**, the temperature detecting resistance wire (the first temperature detecting resistance element) **12b** and the temperature detecting resistance wire (the second temperature detecting resistance element) **12d** spaced along the liquid distribution direction FD and formed on the detection surface **12A**.

[0049] The detection surface **12A** is a flatly formed surface extending along the axis X. The heating resistance wire **12a**, and the temperature detecting resistance wire **12b**, **12c**, **12d**, **12e** are each formed of a metal film, such as of platinum, evaporated onto the glass substrate.

[0050] The liquid flowing through the measurement tube **11** flows along the axis X in a flow direction FD from the left toward the right in FIG. **4**. Accordingly, when the heating resistance wire **12a** is heated momentarily, the heated liquid flows along the axis X to a position of the temperature detecting resistance wire **12b** and then to a position of the temperature detecting resistance wire **12d**. The control substrate **20** detects voltage signals corresponding to electrical resistance values of the temperature detecting resistance wire **12b** and the temperature detecting resistance wire **12d** that change with temperature, to measure temperatures of the temperature detecting resistance wire **12b** and the temperature detecting resistance wire **12d**.

[0051] A middle position of the sensor substrate **12** in a direction of an axis X is on an outlet **11b** side of an intermediate position at an equal distance from both the inlet **11a** and the outlet **11b** of the measurement tube **11**. Furthermore, a position where the heating resistance wire **12a** is disposed in the sensor substrate **12** is on the outlet **11b** side of an intermediate position at an equal distance from both an inlet **11a** side end portion of the sensor substrate **12** and an outlet **11b** side end portion of the sensor substrate **12**.

[0052] A distance from the inlet **11a** of the measurement tube **11** to the heating resistance wire **12a** on the axis X is longer than a distance from the outlet **11b** of the measurement tube **11** to the heating resistance wire **12a** on the axis X. Consequently, the distance from the inlet **11a** of the measurement tube **11** to the heating resistance wire **12a** can be increased, and turbulence or the like of the liquid flowing into the inlet **11a** of the measurement tube **11** can be sufficiently reduced before the liquid is heated.

[0053] As shown in FIG. **4**, in the detection surface **12A**, a wiring pattern **12f** connected to one end

of the heating resistance wire **12a** and a wiring pattern **12g** connected to the other end of the heating resistance wire **12a** are formed. Also, in the detection surface **12A**, a wiring pattern **12h** connected to one end of the temperature detecting resistance wire **12b**, a wiring pattern **12i** connected to the other end of the temperature detecting resistance wire **12b** and a wiring pattern **12j** connected to one end of the temperature detecting resistance wire **12d** are formed. The other end of the temperature detecting resistance wire **12d** is connected to the wiring pattern **12i**.

[0054] Also, in the detection surface **12A**, a wiring pattern **12k** connected to one end of the temperature detecting resistance wire **12c**, a wiring pattern **12l** connected to the other end of the temperature detecting resistance wire **12c** and a wiring pattern **12m** connected to one end of the temperature detecting resistance wire **12e** are formed. The other end of the temperature detecting resistance wire **12e** is connected to the wiring pattern **12l**. The wiring patterns **12f**, **12g**, **12h**, **12i**, **12j**, **12k**, **12l**, **12m** are each formed of a metal film, such as of platinum, evaporated onto the glass substrate.

[0055] End portions of the wiring patterns **12f**, **12g**, **12h**, **12i**, **12j**, **12k**, **12l**, **12m** are joined to metal wiring patterns **60f**, **60g**, **60h**, **60i**, **60j**, **60k**, **60l**, **60m** arranged in the flexible substrate (the external connection terminal) **60** formed of film resin, respectively. Each of the wiring patterns **60f**, **60g**, **60h**, **60i**, **60j**, **60k**, **60l**, **60m** of the flexible substrate **60** is electrically connected to the control substrate **20**.

[0056] As shown in FIG. 3, the measurement tube **11** has an almost circular upper side of a cross section defined by a plane orthogonal to the axis X at a position to which the sensor substrate **12** is bonded. A part of an outer circumferential surface of the measurement tube **11** that is disposed as opposed to the detection surface **12A** of the sensor substrate **12** is the flat surface **11c** that is flat. On the other hand, the measurement tube **11** has a circular cross section defined by the plane orthogonal to the axis X at a position of the tube to which the sensor substrate **12** is not bonded. The flat surface **11c** of the measurement tube **11** is disposed as opposed to the detection surface **12A** of the sensor substrate **12**. The detection surface **12A** is bonded with the adhesive **80** to the flat surface **11c** of the measurement tube **11** along the axis X.

[0057] A distance D1 from the detection surface **12A** of the sensor substrate **12** to an inner circumferential surface **10d** of the internal flow passage **10c** is shorter than a distance D2 from a top **11d** of the measurement tube **11** to the inner circumferential surface **10d** of the internal flow passage **10c**. Thus, the distance D1 from the detection surface **12A** of the sensor substrate **12** to the inner circumferential surface **10d** of the internal flow passage **10c** is shorter than the distance D2, to improve thermal conductivity from the heating resistance wire **12a** to the liquid and to improve properties of the temperature detecting resistance wire **12b** and the temperature detecting resistance wire **12d** to detect the temperatures. The distance D1 is preferably 0.2 mm or less.

[0058] As shown in FIG. 2, the inlet-side body **16** receives the inlet **11a** of the measurement tube **11** and has inside a connection flow passage **16a** that has a circular cross section. The outlet-side body **17** receives the outlet **11b** of the measurement tube **11** and has inside a connection flow passage **17a** that has a circular cross section.

[0059] The nuts **15** include the inlet-side nut **15a** attached to the inlet-side body **16** and the outlet-side nut **15b** attached to the outlet-side body **17**. The inlet-side nut **15a** is a cylindrical member fitted along the outer circumferential surface of the measurement tube **11** to be closer to the outlet **11b** than the inlet-side body **16**.

[0060] Next, the control substrate **20** will be described in detail with reference to FIG. 5. FIG. 5 is a block diagram illustrating a control configuration of the control substrate **20** illustrated in FIG. 1. As illustrated in FIG. 5, the control substrate **20** has a first voltage detection unit **21**, a second voltage detection unit **22**, a third voltage detection unit **23**, a fourth voltage detection unit **24**, a voltage output unit **25**, a time measurement unit **26**, and a calculation unit **27**.

[0061] The voltage output unit **25** outputs a voltage signal used for heating a heating resistance wire **12a** to the sensor substrate **12** via the wiring patterns **60f**, **60g** of the flexible substrate **60**. The

voltage output unit **25** outputs a voltage signal to the heating resistance wire **12a** so as to periodically repeat a heating period for heating the heating resistance wire **12a** and a non-heating period for not heating the heating resistance wire **12a**.

[0062] For example, the voltage output unit **25** outputs a voltage signal so that the heating period is shorter than the non-heating period. The heating period is set to a ratio less than 0.5 in a single cycle that is a sum of a heating period and a non-heating period. The ratio in a single cycle of the heating period may be set to be less than 0.4.

[0063] The first voltage detection unit **21** determines a first output voltage V_{o1} in accordance with the temperature of the temperature detecting resistance wire **12b**. The first voltage detection unit **21** supplies current at a constant current value to the temperature detecting resistance wire **12b** and determines the first output voltage V_{o1} in accordance with the resistance value of the temperature detecting resistance wire **12b** that varies with the temperature.

[0064] The second voltage detection unit **22** determines a second output voltage V_{o2} in accordance with the temperature of the temperature detecting resistance wire **12d**. The second voltage detection unit **22** supplies current at a constant current value to the temperature detecting resistance wire **12d** and determines the second output voltage V_{o2} in accordance with the resistance value of the temperature detecting resistance wire **12d** that varies with the temperature.

[0065] The third voltage detection unit **23** determines a third output voltage V_{o3} in accordance with the temperature of the temperature detecting resistance wire **12c**. The third voltage detection unit **23** supplies current at a constant current value to the temperature detecting resistance wire **12c** and determines the third output voltage V_{o3} in accordance with the resistance value of the temperature detecting resistance wire **12c** that varies with the temperature.

[0066] The fourth voltage detection unit **24** determines a fourth output voltage V_{o4} in accordance with the temperature of the temperature detecting resistance wire **12e**. The fourth voltage detection unit **24** supplies current at a constant current value to the temperature detecting resistance wire **12e** and determines the fourth output voltage V_{o4} in accordance with the resistance value of the temperature detecting resistance wire **12e** that varies with the temperature.

[0067] The time measurement unit **26** measures a first reach time T_1 , which is taken by a liquid heated by the heating resistance wire **12a** to reach the first temperature detecting resistance wire **12b**, based on a voltage signal output by the voltage output unit **25** and the first output voltage V_{o1} determined by the first voltage detection unit **21**. Further, the time measurement unit **26** measures a second reach time T_2 , which is taken by a liquid heated by the heating resistance wire **12a** to reach the second temperature detecting resistance wire **12d**, based on a voltage signal output by the voltage output unit **25** and the second output voltage V_{o2} determined by the second voltage detection unit **22**.

[0068] Note that, when measuring the first reach time T_1 , it is preferable for the time measurement unit **26** to measure the first reach time T_1 based on a first differential voltage $DV_1(V_{o1}-V_{o3})$ obtained by subtracting the third output voltage V_{o3} determined by the third voltage detection unit **23** from the first output voltage V_{o1} determined by the first voltage detection unit **21**. This can eliminate the effect due to heat that is not transferred to the liquid but transferred to the temperature detecting resistance wire **12c** via the measurement tube **11** out of the heat transferred to the measurement tube **11** by the heating resistance wire **12a** and thus can improve the measurement accuracy.

[0069] The temperature detecting resistance wire **12b** and the temperature detecting resistance wire **12c** are arranged at the same distance from the heating resistance wire **12a**. The third output voltage V_{o3} determined by the temperature detecting resistance wire **12c** upstream in the flow direction FD of the heating resistance wire **12a** corresponds to the heat that is not transferred to the liquid but transferred to the temperature detecting resistance wire **12b** via the measurement tube **11** out of the heat transferred to the measurement tube **11** by the heating resistance wire **12a**.

Therefore, by measuring the first reach time T_1 based on the first differential voltage DV_1 , it is

possible to eliminate the effect due to the heat transferred to the temperature detecting resistance wire **12c** via the measurement tube **11**.

[0070] Similarly, when measuring the second reach time **T2**, it is preferable for the time measurement unit **26** to measure the second reach time **T2** based on a second differential voltage **DV2** (**Vo2-Vo4**) obtained by subtracting the fourth output voltage **Vo4** determined by the fourth voltage detection unit **24** from the second output voltage **Vo2** determined by the second voltage detection unit **22**. This can eliminate the effect due to heat that is not transferred to the liquid but transferred to the temperature detecting resistance wire **12d** via the measurement tube **11** out of the heat transferred to the measurement tube **11** by the heating resistance wire **12a** and thus can improve the measurement accuracy.

[0071] The temperature detecting resistance wire **12d** and the temperature detecting resistance wire **12e** are arranged at the same distance from the heating resistance wire **12a**. The fourth output voltage **Vo4** determined by the temperature detecting resistance wire **12e** upstream in the flow direction **FD** of the heating resistance wire **12a** corresponds to the heat that is not transferred to the liquid but transferred to the temperature detecting resistance wire **12d** via the measurement tube **11** out of the heat transferred to the measurement tube **11** by the heating resistance wire **12a**.

Therefore, by measuring the second reach time **T2** based on the second differential voltage **DV2**, it is possible to eliminate the effect due to the heat transferred to the temperature detecting resistance wire **12d** via the measurement tube **11**.

[0072] For example, the time measurement unit **26** performs Fourier transform on time-series signals of the first differential voltage **DV1** and calculates the first reach time **T1** based on a coefficient of the fundamental wave (a sinewave of the same frequency as the voltage signal output by the voltage output unit **25**) resulted after the Fourier transform. Further, for example, the time measurement unit **26** performs Fourier transform on time-series signals of the second differential voltage **DV2** and calculates the second reach time **T2** based on a coefficient of the fundamental wave resulted after the Fourier transform.

[0073] For example, the time measurement unit **26** performs Fourier transform on time-series signals of the first differential voltage **DV1** and calculates the amplitude **VA1** of the fundamental wave resulted after the Fourier transform. The amplitude **VA1** is a value in accordance with a heat quantity of a liquid transferred to the temperature detecting resistance wire **12b**. Further, for example, the time measurement unit **26** performs Fourier transform on time-series signals of the second differential voltage **DV2** and calculates the amplitude **VA2** of the fundamental wave resulted after the Fourier transform. The amplitude **VA2** is a value in accordance with a heat quantity of a liquid transferred to the temperature detecting resistance wire **12d**.

[0074] The calculation unit **27** calculates the liquid flow rate of a liquid flowing through the measurement tube **11** based on the first reach time **T1** and the second reach time **T2** measured by the time measurement unit **26**. The calculation unit **27** switches the calculation formula used for calculation of the liquid flow rate in accordance with the flow rate that is currently being calculated by the calculation unit **27**. The operation performed by the calculation unit **27** to switch the calculation formula in accordance with the flow rate to calculate the liquid flow rate will be described below.

[0075] FIG. **6** is a graph illustrating a relationship between the liquid flow rate calculated by the calculation unit **27** and the amplitudes of the first differential voltage **DV1** and the second differential voltage **DV2**. In FIG. **6**, the first region **AR1** is a region in which the flow rate is greater than or equal to 4 [$\mu\text{L}/\text{min}$] and less than a first predetermined flow rate **PFL1** [$\mu\text{L}/\text{min}$], the second region **AR2** is a region in which the flow rate is greater than or equal to the first predetermined flow rate **PFL1** [$\mu\text{L}/\text{min}$] and less than a second predetermined flow rate **PFL2** [$\mu\text{L}/\text{min}$], and the third region **AR3** is a region in which the flow rate is greater than or equal to the second predetermined flow rate **PFL2** [$\mu\text{L}/\text{min}$]. For example, the first predetermined flow rate **PFL1** is 8 $\mu\text{L}/\text{min}$, and the second predetermined flow rate **PFL2** is 20 $\mu\text{L}/\text{min}$. The first predetermined flow

rate PFL1 and the second predetermined flow rate PFL2 may be set to any other values.

[0076] Note that the flow rate in the graph illustrated in FIG. 6 differs from the flow rate calculated by the calculation unit 27 of the present embodiment and is a flow rate measured by a different type of a calibrating flow meter (for example, a Coriolis flow meter) that can accurately measure the liquid flow rate of a liquid flowing through the internal flow passage 10c in any flow rate region of the first region AR1, the second region AR2, and the third region AR3.

[0077] As illustrated in FIG. 6, in the third region AR3 (in particular, the region of about 50 $\mu\text{L}/\text{min}$ or greater), the difference is small between the amplitude VA1 of the fundamental wave of the first differential voltage DV1 resulted after Fourier transform and the amplitude VA2 of the fundamental wave of the second differential voltage DV2 resulted after Fourier transform. In the first region AR1 and the second region AR2, however, the amplitude VA1 is greater than the amplitude VA2, and the difference between the amplitude VA1 and the amplitude VA2 increases as the liquid flow rate decreases.

[0078] As described above, the reason for the increase in the difference between the amplitude VA1 and the amplitude VA2 in the first region AR1 and the second region AR2 is that, when the liquid flow rate is relatively small, heat applied to the liquid from the heating resistance wire 12a will be partially transferred to the measurement tube 11 or the liquid itself before reaching the temperature detecting resistance wire 12d. Accordingly, in the present embodiment, the calculation unit 27 switches the calculation formula used for calculation of the flow rate in accordance with whether the flow rate to be calculated by the calculation unit 27 is in the first region AR1, the second region AR2, or the third region AR3.

[0079] Next, a flow rate calculation method for calculating the liquid flow rate performed by the thermal flow meter 100 of the present embodiment will be described. FIG. 7 is a flowchart illustrating the flow rate calculation method for calculating the liquid flow rate performed by the thermal flow meter 100 according to one embodiment of the present disclosure. Each process of the flowchart illustrated in FIG. 7 is performed by a control program stored in the control substrate 20.

[0080] In step S101, the voltage output unit 25 of the control substrate 20 outputs a voltage signal to the heating resistance wire 12a to heat a liquid flowing through the measurement tube 11. The voltage output unit 25 outputs the voltage signal to the heating resistance wire 12a so as to periodically repeat a heating period for heating the heating resistance wire 12a and a non-heating period for not heating the heating resistance wire 12a.

[0081] In step S102, the first voltage detection unit 21 of the control substrate 20 determines the first output voltage Vo1 between the wiring pattern 60h and the wiring pattern 60i connected to the temperature detecting resistance wire 12b. The second voltage detection unit 22 of the control substrate 20 determines the second output voltage Vo2 between the wiring pattern 60j and the wiring pattern 60i connected to the temperature detecting resistance wire 12d. The third voltage detection unit 23 of the control substrate 20 determines the third output voltage Vo3 between the wiring pattern 60k and the wiring pattern 60l connected to the temperature detecting resistance wire 12c. The fourth voltage detection unit 24 of the control substrate 20 determines the fourth output voltage Vo4 between a wiring pattern 60m and the wiring pattern 60l connected to the temperature detecting resistance wire 12e.

[0082] In step S103, the time measurement unit 26 of the control substrate 20 performs Fourier transform on time-series signals of the first differential voltage DV1 obtained by subtracting the third output voltage Vo3 from the first output voltage Vo1 and calculates the first reach time T1 based on a coefficient of the fundamental wave (a sinewave of the same frequency as the voltage signal output by the voltage output unit 25) resulted after the Fourier transform.

[0083] In step S104, the calculation unit 27 of the control substrate 20 calculates a first flow rate FL1 of the liquid based on the first reach time T1 by using the following calculation formula (A).

[00001] $FL1 = S \times D1 / T1$ (A)

[0084] Herein, S [m.sup.2] is the sectional area of the internal flow passage **10c** of the measurement tube **11**, and $D1$ [m] is the distance from the heating resistance wire **12a** to the temperature detecting resistance wire **12b**.

[0085] Note that, instead of the above calculation formula (A), the following calculation formula (A1) may be used.

$$[00002] \text{ FL1} = a0 + a1 / T1 + a2 / T1^2 + a3 / T1^3 + a4 / T1^4 \quad (A1)$$

[0086] Herein, $a0$, $a1$, $a2$, $a3$, $a4$ are coefficients, respectively, and are calculated in advance based on the relationship between a plurality of flow rates measured by a calibrating flow meter and the first reach time $T1$. The above calculation formula (A1) is approximated by the first flow rate $FL1$ by a fourth order polynomial of the first reach time $T1$, however, may be approximated by a second order, third order, or fifth order or higher polynomial.

[0087] In step **S105**, the time measurement unit **26** of the control substrate **20** performs Fourier transform on time-series signals of the second differential voltage $DV2$ obtained by subtracting the fourth output voltage $Vo4$ from the second output voltage $Vo2$ and calculates the second reach time $T2$ based on a coefficient of the fundamental wave (a sinewave of the same frequency as the voltage signal output by the voltage output unit **25**) resulted after the Fourier transform.

[0088] In step **S106**, the calculation unit **27** of the control substrate **20** calculates a second flow rate $FL2$ of the liquid based on the second reach time $T2$ by using the following calculation formula (B).

$$[00003] \text{ FL2} = S \times D2 / T2 \quad (B)$$

[0089] Herein, $D2$ [m] is the distance from the heating resistance wire **12a** to the temperature detecting resistance wire **12d**.

[0090] Note that, instead of the above calculation formula (B), the following calculation formula (B1) may be used.

$$[00004] \text{ FL2} = b0 + b1 / T2 + b2 / T2^2 + b3 / T2^3 + b4 / T2^4 \quad (B1)$$

[0091] Herein, $b0$, $b1$, $b2$, $b3$, $b4$ are coefficients, respectively, and are calculated in advance based on the relationship between a plurality of flow rates measured by a calibrating flow meter and the second reach time $T2$. The above calculation formula (B1) is approximated by the second flow rate $FL2$ by a fourth order polynomial of the second reach time $T2$, however, may be approximated by a second order, third order, or fifth order or higher polynomial.

[0092] In step **S107**, the calculation unit **27** of the control substrate **20** determines whether or not the first flow rate $FL1$ calculated in step **S104** is less than or equal to the first predetermined flow rate $PFL1$, and the calculation unit **27** proceeds with the process to step **S108** if the determination is YES or proceeds with the process to step **S109** if the determination is NO.

[0093] In step **S108**, step **S110**, and step **S111**, the calculation unit **27** calculates the liquid flow rate FL of the liquid by the following calculation formula (C), respectively.

$$[00005] \text{ FL} = C1 \times FL1 + C2 \times FL2 \quad (C)$$

[0094] Herein, the first coefficient $C1$ and the second coefficient $C2$ are values the sum of which is 1 and, in calculation of the flow rate FL , indicate weights on the first flow rate $FL1$ calculated based on the first output voltage $Vo1$ output from the temperature detecting resistance wire **12b** and the second flow rate $FL2$ calculated based on the second output voltage $Vo2$ output from the temperature detecting resistance wire **12d**.

[0095] In step **S108**, the calculation unit **27** of the control substrate **20** calculates the liquid flow rate FL by the following calculation formula (1).

$$[00006] \text{ FL} = FL1 \quad (1)$$

[0096] The above calculation formula (1) is obtained when the first coefficient $C1$ is 1 and the second coefficient $C2$ is 0 in the calculation formula (C). The calculation formula (1) is a formula in which the weight on the flow rate that is calculated based on the second output voltage $Vo2$

output from the temperature detecting resistance wire **12d** is 0.

[0097] In step **S109**, the calculation unit **27** of the control substrate **20** determines whether or not the latest calculated liquid flow rate is less than or equal to the second predetermined flow rate **PFL2**, and the calculation unit **27** proceeds with the process to step **S110** if the determination is YES or proceeds with the process to step **S111** if the determination is NO.

[0098] In step **S110**, the calculation unit **27** of the control substrate **20** calculates the liquid flow rate **FL** by the following calculation formula (2).

$$[00007] \text{ FL} = \text{C1} \times \text{FL1} + \text{C2} \times \text{FL2} \quad (2)$$

[0099] The above calculation formula (2) is obtained when respective values of the first coefficient **C1** and the second coefficient **C2** are greater than 0 and less than 1 in the calculation formula (A). The calculation formula (2) is a formula that applies weights to the flow rate calculated based on the first output voltage **Vo1** output from the temperature detecting resistance wire **12b** and the flow rate calculated based on the second output voltage **Vo2** output from the temperature detecting resistance wire **12d**, respectively.

[0100] In step **S111**, the calculation unit **27** of the control substrate **20** calculates the liquid flow rate **FL** by the following calculation formula (3).

$$[00008] \text{ FL} = 0.5 \times \text{FL1} + 0.5 \times \text{FL2} \quad (3)$$

[0101] The above calculation formula (3) is obtained when the first coefficient **C1** and the second coefficient **C2** are 0.5, respectively. The calculation formula (3) is a formula that applies an equal weight to the flow rate calculated based on the first output voltage **Vo1** output from the temperature detecting resistance wire **12b** and the flow rate calculated based on the second output voltage **Vo2** output from the temperature detecting resistance wire **12d**, respectively.

[0102] Once calculating the liquid flow rates **FL** in step **S108**, step **S110**, and step **S111**, the control substrate **20** transmits an output signal indicating the calculated flow rate **FL** to an external device via the relay substrate **30** and ends the process of the present flowchart. The control substrate **20** continuously calculates the liquid flow rate **FL** by repeatedly performing the above process of step **S101** to step **S111**.

[0103] FIG. **8** is a graph illustrating a relationship between the liquid flow rate calculated by the calculation unit **27** and the first coefficient **C1** and the second coefficient **C2** used in the calculation formulae (1), (2), and (3). As illustrated in FIG. **8**, when the flow rate calculated by the calculation unit **27** is in the first region **AR1**, the first coefficient **C1** is fixed to 1, and the second coefficient **C2** is fixed to 0. Further, when the flow rate calculated by the calculation unit **27** is in the third region **AR3**, the first coefficient **C1** and the second coefficient **C2** are fixed to 0.5, respectively.

[0104] However, when the flow rate calculated by the calculation unit **27** is in the second region **AR2**, the first coefficient **C1** and the second coefficient **C2** are greater than 0 and less than 1, respectively. The first coefficient **C1** is gradually increased at a constant increase rate as the liquid flow rate calculated by the calculation unit **27** decreases from the second predetermined flow rate **PFL2** (20 $\mu\text{L}/\text{min}$) in FIG. **8**) to the first predetermined flow rate **PFL1** (8 $\mu\text{L}/\text{min}$) in FIG. **8**). The second coefficient **C2** is gradually reduced at a constant reduction rate as the liquid flow rate calculated by the calculation unit **27** decreases from the second predetermined flow rate **PFL2** to the first predetermined flow rate **PFL1**.

[0105] Note that, in FIG. **8**, the first coefficient **C1** and the second coefficient **C2** applied when the flow rate calculated by the calculation unit **27** is in the second region **AR2** may be values of other modified examples. FIG. **9** is a graph illustrating a modified example of the relationship between the liquid flow rate calculated by the calculation unit **27** and the first coefficient **C1** and the second coefficient **C2** used in the calculation formulae. As illustrated in FIG. **9**, when the flow rate calculated by the calculation unit **27** is in the second region **AR2**, the first coefficient **C1** may be increased stepwise as the liquid flow rate calculated by the calculation unit **27** decreases from the second predetermined flow rate **PFL2** to the first predetermined flow rate **PFL1**. The second

coefficient C2 may be reduced stepwise as the liquid flow rate calculated by the calculation unit 27 decreases from the second predetermined flow rate PFL2 to the first predetermined flow rate PFL1. [0106] As described above, when the first flow rate FL1 is greater than the first predetermined flow rate PFL1, the calculation unit 27 calculates the liquid flow rate based on both the first reach time T1 and the second reach time T2 in step S110 and step S111. However, when the liquid flow rate is less than or equal to the first predetermined flow rate PFL1, the calculation unit 27 calculates the liquid flow rate based on the first reach time T1 without using the second reach time T2.

[0107] When the first flow rate FL1 is greater than the first predetermined flow rate PFL1, the calculation unit 27 calculates the liquid flow rate FL by applying predetermined weights to the first flow rate FL1 calculated based on the first reach time T1 and the second flow rate FL2 calculated based on the second reach time T2, and when the first flow rate FL1 is less than or equal to the first predetermined flow rate PFL1, the calculation unit 27 calculates the liquid flow rate FL by increasing the weight on the first flow rate FL1 to be greater than the predetermined weight.

[0108] Further, when the first flow rate FL1 is greater than the second predetermined flow rate PFL2, which is greater than the first predetermined flow rate PFL1, the calculation unit 27 calculates the liquid flow rate by applying an equal weight to the first flow rate FL1 calculated based on the first reach time T1 and the second flow rate FL2 calculated based on the second reach time T2. However, when the first flow rate FL1 is greater than the first predetermined flow rate PFL1 and less than or equal to the second predetermined flow rate PFL2, the calculation unit 27 calculates the liquid flow rate FL by increasing the weight on the first flow rate FL1 to be greater than the weight on the second flow rate FL2.

[0109] The effects and advantages achieved by the thermal flow meter 100 of the present embodiment described above will be described.

[0110] According to the thermal flow meter 100 of the present embodiment, when the liquid flow rate is greater than the first predetermined flow rate PFL1, the liquid flow rate FL is calculated based on both the first reach time T1 and the second reach time T2. Thus, the error in calculation of the flow rate FL can be reduced compared to a case where the flow rate is calculated without the use of either one of the first reach time T1 and the second reach time T2.

[0111] Further, according to the thermal flow meter 100 of the present embodiment, when the liquid flow rate FL is less than or equal to the first predetermined flow rate PFL1, the liquid flow rate FL is calculated based on the first reach time T1 without the use of the second reach time T2. Thus, the calculation error of the liquid flow rate FL due to heat applied to a liquid from the heating resistance wire 12a being transferred to the measurement tube 11 or the liquid itself before reaching the second temperature detecting resistance wire 12d can be reduced, and the measurable flow rate range can be expanded on the lower limit value side.

[0112] According to the thermal flow meter 100 of the present embodiment, when the liquid flow rate is greater than the second predetermined flow rate PFL2, the liquid flow rate FL is calculated with an equal weight being applied to the first flow rate FL1 calculated based on the first reach time T1 and the second flow rate FL2 calculated based on the second reach time T2. Thus, the error in calculation of the flow rate FL can be reduced compared to a case where the flow rate FL is calculated based only on either one of the first flow rate FL1 calculated based on the first reach time T1 and the second flow rate FL2 calculated based on the second reach time T2.

[0113] Further, according to the thermal flow meter 100 of the present embodiment, when the liquid flow rate is greater than the first predetermined flow rate PFL1 and less than or equal to the second predetermined flow rate PFL2, the liquid flow rate FL is calculated with the weight on the first flow rate FL1 being increased to be greater than the weight on the second flow rate FL2. Thus, the calculation error of the liquid flow rate due to heat applied to a liquid from the heating resistance wire 12a being transferred to the measurement tube 11 or the liquid itself before reaching the second temperature detecting resistance wire 12d can be reduced.

[0114] According to the thermal flow meter 100 of the present embodiment, the first coefficient C1

is increased and the second coefficient C2 is reduced as the liquid flow rate FL calculated by the calculation unit 27 decreases and is closer to the first predetermined flow rate PFL1 from the second predetermined flow rate PFL2, and thus, the calculation error of the liquid flow rate due to heat applied to a liquid from the heating resistance wire 12a being transferred to the measurement tube 11 or the liquid itself before reaching the second temperature detecting resistance wire 12d can be reduced in accordance with a reduction in the flow rate FL.

[0115] According to the thermal flow meter 100 of the present embodiment, the first coefficient C1 is gradually increased at a constant increase rate and the second coefficient C2 is gradually reduced at a constant reduction rate as the liquid flow rate FL calculated by the calculation unit 27 decreases and is closer to the first predetermined flow rate PFL1 from the second predetermined flow rate PFL2, and thus, fluctuation in the calculation result of the flow rate FL can be suitably suppressed.

Claims

1. A thermal flow meter comprising: a measurement tube having an inlet and an outlet, a liquid flowing in the inlet and the liquid, which flowed in from the inlet, flowing out of the outlet, and having an internal flow passage extending along an axis formed in the measurement tube; a temperature detection substrate having a heating resistance element, a first temperature detecting resistance element, and a second temperature detecting resistance element that are formed on a detection surface, each extend in a flow direction from the inlet to the outlet, and are spaced apart from each other; a voltage output unit configured to output a voltage signal to the temperature detection substrate, the voltage signal being for heating the heating resistance element; a voltage detection unit configured to determine a first output voltage in accordance with a temperature of the first temperature detecting resistance element and a second output voltage in accordance with a temperature of the second temperature detecting resistance element; a time measurement unit configured to measure a first reach time based on the voltage signal and the first output voltage and measure a second reach time based on the voltage signal and the second output voltage, the first reach time being a time taken by a liquid heated by the heating resistance element to reach the first temperature detecting resistance element, and the second reach time being a time taken by a liquid heated by the heating resistance element to reach the second temperature detecting resistance element; and a calculation unit configured to calculate a liquid flow rate of a liquid flowing through the measurement tube based on the first reach time and the second reach time, wherein the calculation unit calculates the liquid flow rate based on both the first reach time and the second reach time when the liquid flow rate is greater than a first predetermined flow rate, and the calculation unit calculates the liquid flow rate based on the first reach time without using the second reach time when the liquid flow rate is less than or equal to the first predetermined flow rate.

2. The thermal flow meter according to claim 1, wherein the calculation unit calculates a liquid flow rate by applying an equal weight to a first flow rate calculated based on the first reach time and to a second flow rate calculated based on the second reach time when the liquid flow rate is greater than a second predetermined flow rate, which is greater than the first predetermined flow rate, and the calculation unit calculates a liquid flow rate by increasing a weight on the first flow rate to be greater than a weight on the second flow rate when the liquid flow rate is greater than the first predetermined flow rate and less than or equal to the second predetermined flow rate.

3. The thermal flow meter according to claim 2, wherein as the liquid flow rate calculated by the calculation unit decreases from the second predetermined flow rate to the first predetermined flow rate, the calculation unit calculates the liquid flow rate by gradually increasing the weight on the first flow rate at a constant increase rate and by gradually reducing the weight on the second flow rate at a constant reduction rate.

4. The thermal flow meter according to claim 1, wherein the calculation unit calculates the liquid

flow rate by applying predetermined weights to a first flow rate calculated based on the first reach time and to a second flow rate calculated based on the second reach time when a liquid flow rate is greater than the first predetermined flow rate, and the calculation unit calculates the liquid flow rate by increasing a weight on the first flow rate to be greater than the predetermined weights when the liquid flow rate is less than or equal to the first predetermined flow rate.

5. A flow rate calculation method for calculating a liquid flow rate by a thermal flow meter, wherein the thermal flow meter comprises a measurement tube having an inlet and an outlet, a liquid flowing in the inlet and the liquid, which flowed in from the inlet, flowing out of the outlet, and having an internal flow passage extending along an axis formed in the measurement tube, a temperature detection substrate having a heating resistance element, a first temperature detecting resistance element, and a second temperature detecting resistance element that are formed on a detection surface, each extend in a flow direction from the inlet to the outlet, and are spaced apart from each other, a voltage output unit configured to output a voltage signal to the temperature detection substrate, the voltage signal being for heating the heating resistance element, and a voltage detection unit configured to determine a first output voltage in accordance with a temperature of the first temperature detecting resistance element and a second output voltage in accordance with a temperature of the second temperature detecting resistance element, the flow rate calculation method comprising: a time measurement step of measuring a first reach time based on the voltage signal and the first output voltage and measuring a second reach time based on the voltage signal and the second output voltage, the first reach time being a time taken by a liquid heated by the heating resistance element to reach the first temperature detecting resistance element, and the second reach time being a time taken by a liquid heated by the heating resistance element to reach the second temperature detecting resistance element; and a calculation step of calculating a liquid flow rate of a liquid flowing through the measurement tube based on the first reach time and the second reach time, wherein the calculation step includes calculating the liquid flow rate based on both the first reach time and the second reach time when the liquid flow rate is greater than a first predetermined flow rate and calculating the liquid flow rate based on the first reach time without using the second reach time when the liquid flow rate is less than or equal to the first predetermined flow rate.

6. The flow rate calculation method according to claim 5, wherein the calculation step includes calculating the liquid flow rate by applying predetermined weights to a first flow rate calculated based on the first reach time and to a second flow rate calculated based on the second reach time when a liquid flow rate is greater than the first predetermined flow rate and calculating the liquid flow rate by increasing a weight on the first flow rate to be greater than the predetermined weights when the liquid flow rate is less than or equal to the first predetermined flow rate.
